



USER MANUAL
DIRT MINI STOCK



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DEAR iRACING USER,

Congratulations on your purchase of the Mini Stock! From all of us at iRacing, we appreciate your support and your commitment to our product. We aim to deliver the ultimate sim racing experience, and we hope that you'll find plenty of excitement with us behind the wheel of your new car!

The following guide explains how to get the most out of your new car, from how to adjust its settings off of the track to what you'll see inside of the cockpit while driving. We hope that you'll find it useful in getting up to speed.

Thanks again for your purchase, and we'll see you on the track!



CHASSIS

STRUT FRONT, 4-LINK REAR SUSPENSION



LENGTH
4521 mm
178 in

WIDTH
1829 mm
72 in

WHEELBASE
2540 mm
100 in

DRY WEIGHT
975 kg
2150 lbs

WET WEIGHT
WITH DRIVER
1067 kg
2350 lbs

POWER UNIT

INLINE 4-CYLINDER 2V OHC



DISPLACEMENT
2.3 Liters
140 CID

RPM LIMIT
8500 RPM

TORQUE
200 lb-ft
260 Nm

POWER
250 bhp
195 kW



INTRODUCTION

The information found in this guide is intended to provide a deeper understanding of the chassis setup adjustments available in the garage, so that you may use the garage to tune the chassis setup to your preference.

Before diving into chassis adjustments, though, it is best to become familiar with the car and track. To that end, we have provided baseline setups for each track commonly raced by these cars. To access the baseline setups, simply open the Garage, click iRacing Setups, and select the appropriate setup for your track of choice. If you are driving a track for which a dedicated baseline setup is not included, you may select a setup for a similar track to use as your baseline.

After you have selected an appropriate setup, get on track and focus on making smooth and consistent laps, identifying the proper racing line and experiencing tire wear and handling trends over a number of laps.

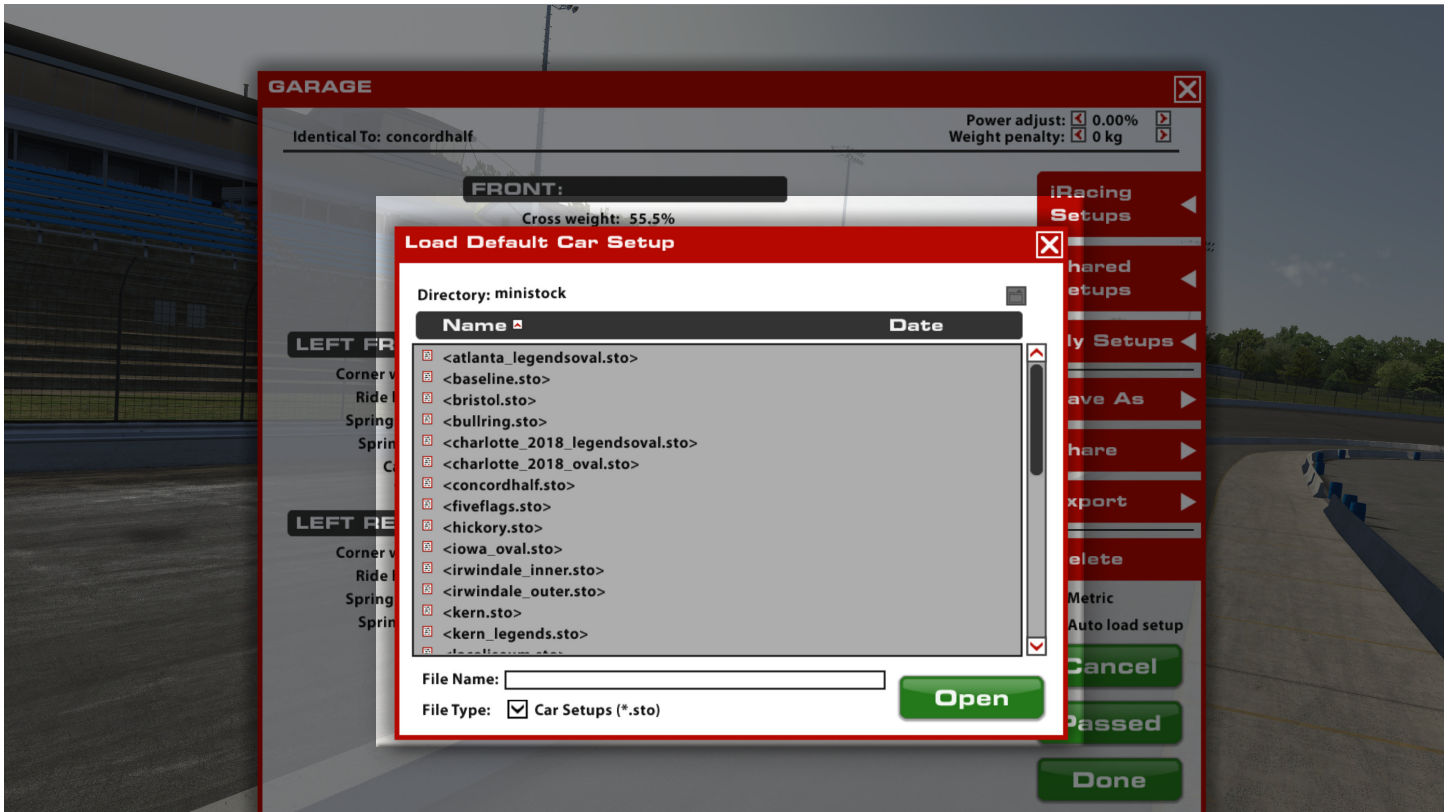
Once you are confident that you are nearing your driving potential with the included baseline setups, read on to begin tuning the car to your handling preferences.

GETTING STARTED



Once you load into the car, select 1st gear and give it a bit of throttle to get underway. The Micro Sprints have five forward gears and a reverse gear but doesn't require a clutch for upshifts or downshifts, with upshifting requiring a quick lift off the throttle and downshifting requiring a small blip of throttle.

LOADING AN iRACING SETUP



When you first load into a session, the iRacing Baseline setup will be automatically loaded onto the car. If you would like to try any of the other iRacing pre-built options, you may select it by going to Garage > iRacing Setups > and then selecting another option that fits your needs. If you would like to customize the setup, simply make the changes in the garage that you would like to update and click apply.

If you would like to save your setup for future use click “Save As” on the right to name and save the changes. To access all of your personally saved setups, click “My Setups” on the right side of the garage. If you would like to share a setup with another driver or everyone in a session, you can select “Share” on the right side of the garage to do so. If a driver is trying to share a setup with you, you will find it under “Shared Setups” on the right side of the garage as well.

DASH CONFIGURATION



The Mini Stock features three analog gauges with a shift light, from left to right:

Water Temperature	Displays the temperature of the water in the engine coolant system, in °F
Tachometer	Displays Engine RPM in 1000 rpm increments.
Oil Pressure	Displays the pressure in the engine oil system in Pounds-per-Square Inch
Shift Light	A shift light is on the far right of the dash that will light up when the engine is at the ideal RPM for upshifting.

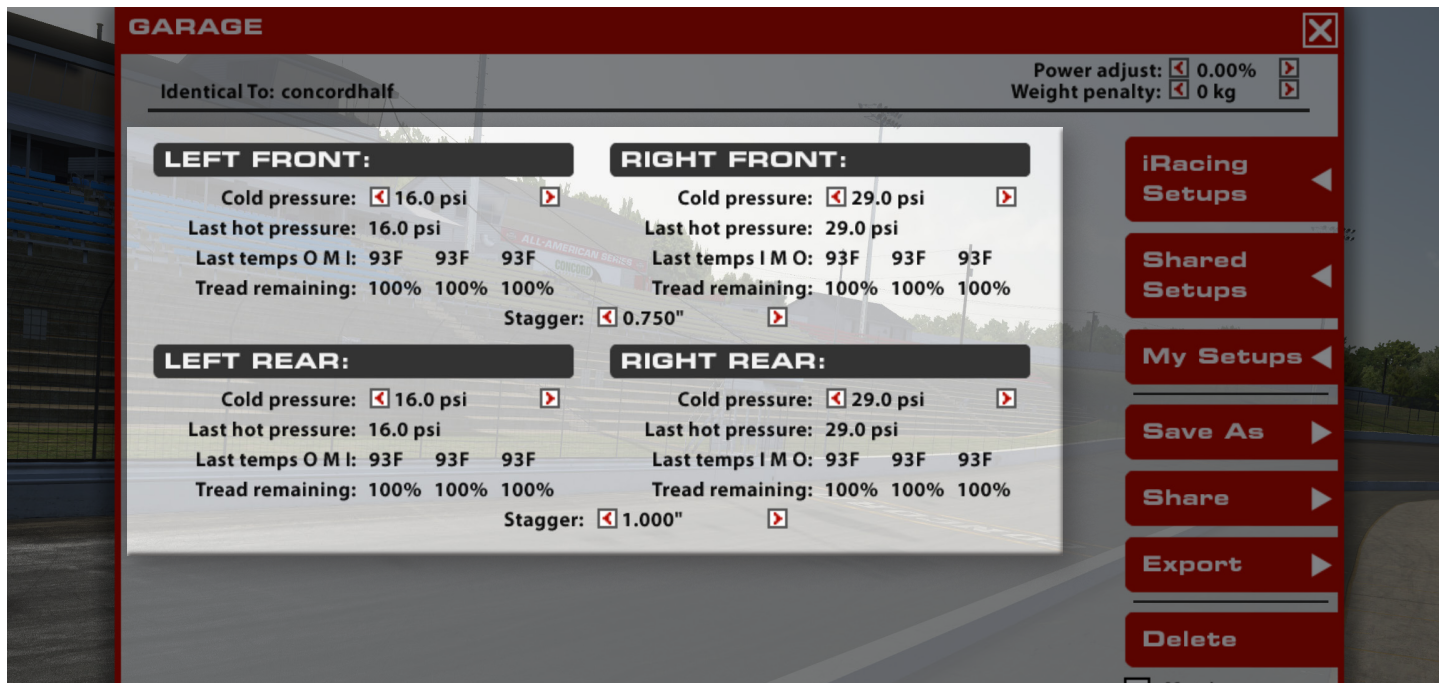
ADVANCED SETUP OPTIONS

This section is aimed toward more advanced users who want to dive deeper into the different aspects of the vehicle's setup. Making adjustments to the following parameters is not required and can lead to significant changes in the way a vehicle handles. It is recommended that any adjustments are made in an incremental fashion and only singular variables are adjusted before testing changes.



TIRES

TIRE DATA



COLD PRESSURE / STARTING PRESSURE

The air pressure in the tires when the car is loaded into the world. Lower pressures will provide more grip but will produce more rolling drag and build temperature faster. Higher pressures will feel slightly more responsive and produce less rolling drag, but will result in less grip. Generally, higher pressures are preferred at tracks where speeds are higher while lower pressures work better at slower tracks where mechanical grip is important.

LAST HOT PRESSURE

When the car returns to the garage after an on-track stint, the tire pressure will be displayed as Hot Pressure. The difference between cold and hot pressure is a good way to see how tires are being loaded and worked while on track. Tires seeing more work will build more pressure, and paying attention to which tires are building more pressure and adjusting cold pressure to compensate can be crucial for optimizing tire performance.

LAST TEMPERATURES

The tire carcass temperatures (measured within the tread) are displayed after the car returns from the track. These temperatures are an effective way to determine how much work or load a given tire is experiencing while on track. Differences between the inner and outer temperatures can be used to tune individual wheel alignment and the center temperatures can be compared to the outer temperatures to help tune tire pressure.

TREAD REMAINING

The amount of tread on the tire, displayed as a percentage of a new tire, is shown below the tire temperatures. These values are good for determining how far a set of tires can go before needing to be replaced, but don't necessarily indicate an under- or over-worked tire in the same way temperatures will.

STAGGER

Stagger is the difference in circumference of the left- and right-side tires with both front and rear stagger available as an adjustment on this car. Increasing the front stagger will increase the size of the right-front tire and add understeer on deceleration and turn-in. Increasing the rear stagger will increase the size of the right-rear tire and add oversteer through the center and on throttle out of the corner. Stagger can be set independently between front and rear to suit various driving styles. For road courses, stagger will default and lock to zero.

CHASSIS

FRONT



CROSS WEIGHT

Cross Weight is the amount of weight on the car's Left-Rear and Right-Front tires relative to the entire weight of the car, displayed in percent. This is adjusted via the corner Spring Perch Offset adjustments as well as ARB preload to a small extent. For an oval car, Cross Weight is one of the most influential settings for grip level while the vehicle is in a turn. Higher Cross Weight values will add weight to the left-rear and right-front, both stabilizing entry and helping drive-off on corner exit. Lower Cross-Weight values will help the vehicle rotate and keep it "free" in the corner to prevent speed from being lost, however too low can result in unstable entry and exit. For this vehicle, excessively high crossweight value can also lead to left-front tire lockups under braking due to reduced load on the left-front wheel.

LEFT ARB END CLEARANCE

The left-side sway bar linkage can be adjusted to either delay bar engagement or apply a static load to the bar. This adjustment directly affects the bar's Preload, outlined below. When "Attach Left Side" is unchecked, having this value positive will require a small amount of front-end travel before the bar is engaged. This can be used to reduce bar load and keep the front end lower on the straights.

ATTACH LEFT SIDE

A quick way to unhook the anti-roll bar to allow for static suspension adjustments without bar twist confusing things; increase link slack and unhook the ARB before making spring/ride height adjustments, attach and re-set End Clearance & Preload when done.

ARB PRELOAD

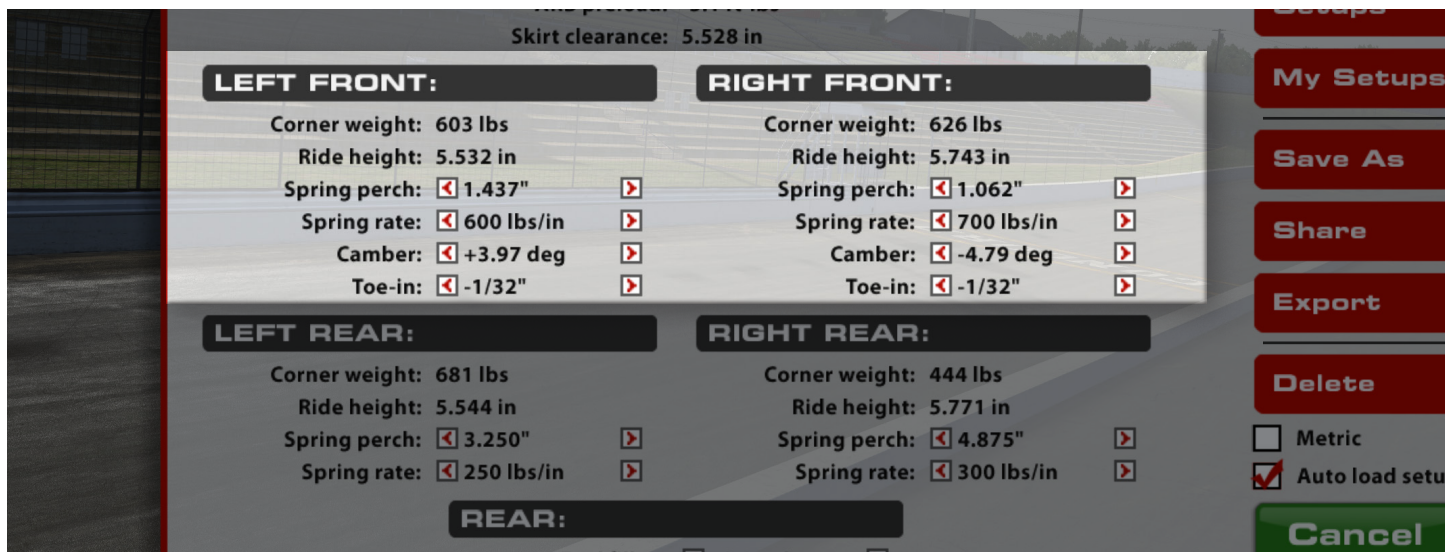
The ARB Preload is the static load in the bar while the vehicle is in the garage. Preload adjustments can be used to alter the dynamic loads in the bar while on track, and can be used to remove or add bar load in the corners and on the straights.

SKIRT CLEARANCE

Distance from the ground to the bottom of the front valence, or skirt. This is a tech inspection measurement and must clear a minimum height to pass tech.



FRONT CORNERS



CORNER WEIGHT

The weight underneath each tire under static conditions in the garage. Correct weight arrangement around the car is crucial for optimizing a car for a given track and conditions. Individual wheel weight adjustments and crossweight adjustments are made via the Shock Collar Offset setting.

RIDE HEIGHT

Distance from ground to a reference point on the chassis. Front heights are measured at the bottom of the chassis frame rail just behind the front wheel wells. Since these values are measured to a specific reference point on the car, these values may not necessarily reflect the vehicle's ground clearance, but instead provide a reliable value for the height of the car off of the race track at static values.

SPRING PERCH

Used to adjust ride height and corner weight, adjusting this setting applies a preload to the spring under static conditions. Decreasing the value increases preload on the spring, adding weight to its corner and increasing the ride height at that corner. Increasing the value does the opposite, reducing height and weight on a given corner. These should be adjusted in pairs (left and right, for example) or with all four spring preload adjustments in the car to prevent crossweight changes while adjusting ride height.

SPRING RATE

Spring Rate changes how stiff the spring is, represented in a force per unit of displacement. Primarily responsible for maintaining ride height and a good attitude under changing wheel loads, stiffer springs will maintain the car's attitude better while sacrificing mechanical grip. Softer springs will deal with bumps better and increase mechanical grip, however this could cause the chassis to pitch and roll too much, causing the car to feel slow to respond to driver inputs.

CAMBER

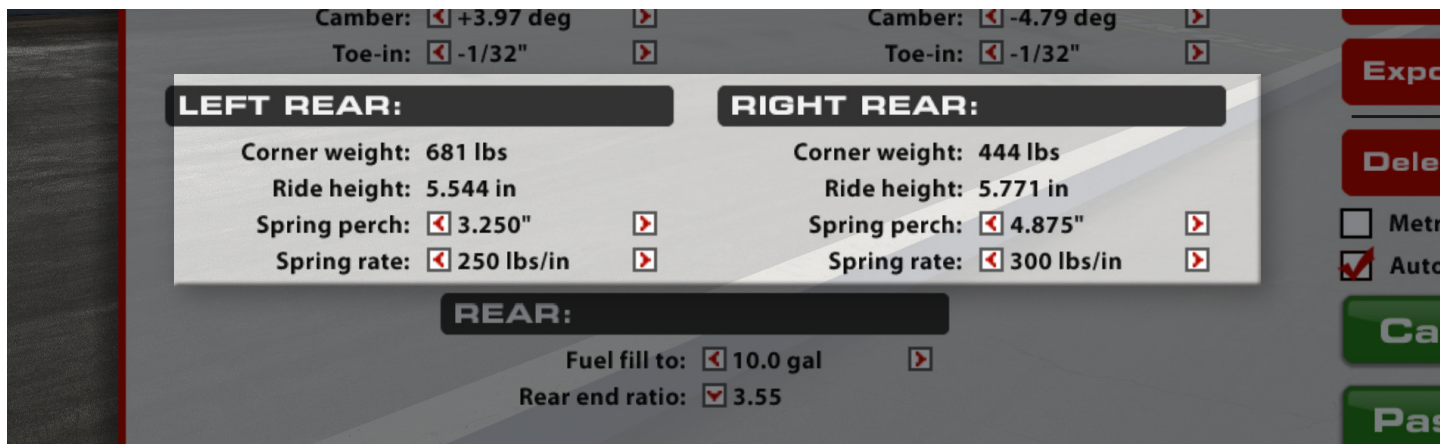
Camber is the vertical angle of the wheel relative to the center of the chassis. Negative camber is when the top of the wheel is closer to the chassis centerline than the bottom of the wheel, positive camber is when the top of the tire is farther out than the bottom. Higher camber angles will increase the cornering force generated by the tire, but will reduce the amount of longitudinal grip the tire will have under braking. Excessive camber values can produce very high cornering forces but will also significantly reduce tire life, so it is important to find a balance between life and performance. For ovals, set the left side positive and the right side negative. For road courses, all four wheels should be set with negative camber.

TOE-IN

Toe is the angle of the wheel, when viewed from above, relative to the centerline of the chassis. Positive toe-in is when the front of the wheel is closer to the centerline than the rear of the wheel, and negative toe-in (toe-out) is when the front of the wheel is farther away from the centerline than the rear of the wheel. More net toe-out will destabilize the car in a straight line and help with turn-in but increases slip angle on the front wheels and can risk over-slipping the front tires, which can result in sudden understeer when cornering.



REAR CORNERS



CORNER WEIGHT

The weight underneath each tire under static conditions in the garage. Correct weight arrangement around the car is crucial for optimizing a car for a given track and conditions. Individual wheel weight adjustments and crossweight adjustments are made via the Shock Collar Offset setting.

RIDE HEIGHT

Distance from ground to a reference point on the chassis. Since these values are measured to a specific reference point on the car, these values may not necessarily reflect the vehicle's ground clearance, but instead provide a reliable value for the height of the car off of the race track at static values.

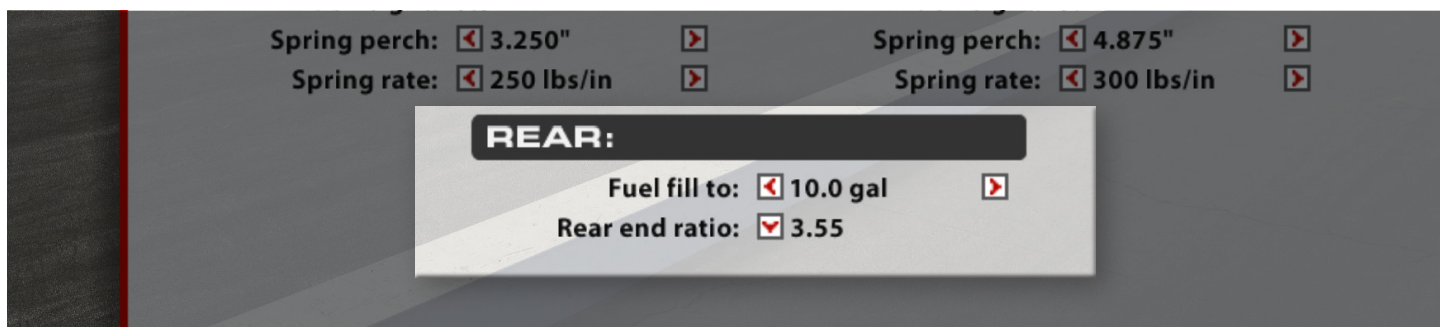
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SPRING RATE

Spring Rate changes how stiff the spring is, represented in force per unit of displacement. Primarily responsible for maintaining ride height and chassis attitude under changing wheel loads, stiffer springs control the chassis attitude better (less roll or pitch change) which is good for alignment control, but mechanical grip is often better with softer springs which allow for more track surface compliance.

REAR



FUEL FILL TO

The Fuel Fill To setting alters the amount of fuel in the fuel tank when the car leaves the garage. Most races will be shorter than what a full tank is capable of, so it is important to get a good fuel reading in practice and carry no more fuel than is necessary for a race to keep overall vehicle weight low.

REAR END RATIO

The Rear End Ratio is the ratio between the driveshaft pinion and the differential ring gear. Higher number values produce better acceleration but reduce top speed, lower number values reduce acceleration but result in a higher top speed.

